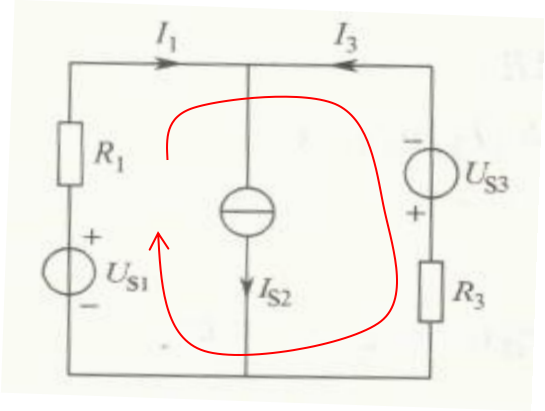
列写支路或回路方程

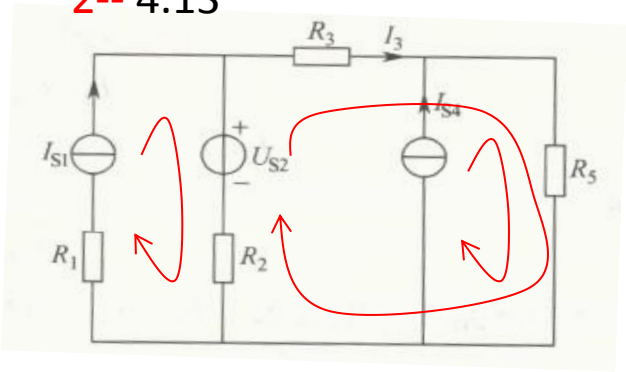
1-- 4.12



$$I_1 + I_3 = I_{S2}$$

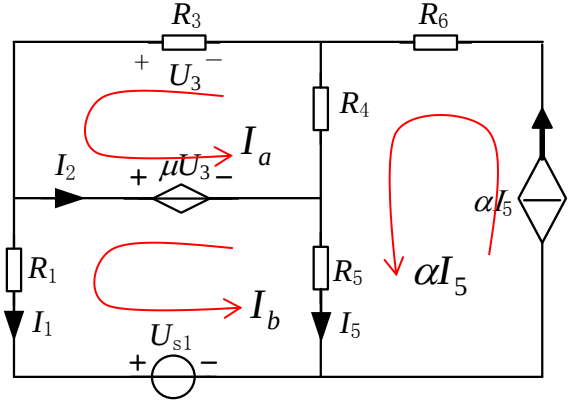
$$R_1 I_1 - I_3 R_3 = U_{S1} + U_{S3}$$

2-- 4.13



$$(R_3 + R_5 + R_2) I_3 - I_{S1} R_2 + I_{S4} R_5 = U_{S2}$$

3-- 4.14



$$(R_3 + R_4) I_a - \alpha I_5 R_4 = -\mu U_3$$

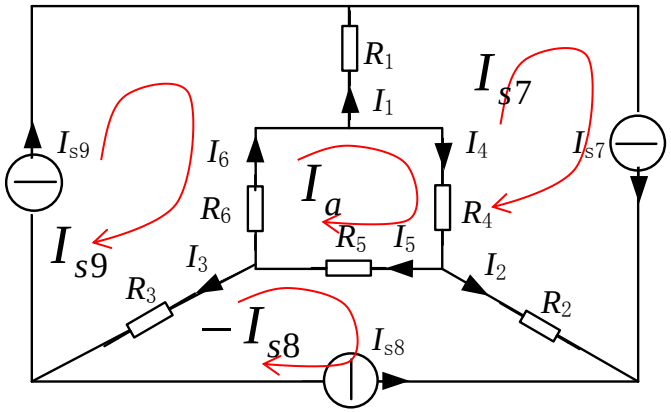
$$(R_1 + R_5) I_b - \alpha I_5 R_5 = \mu U_3 - U_{s1}$$

$$U_3 = -R_3 I_a$$

$$I_5 = \alpha I_5 - I_b$$

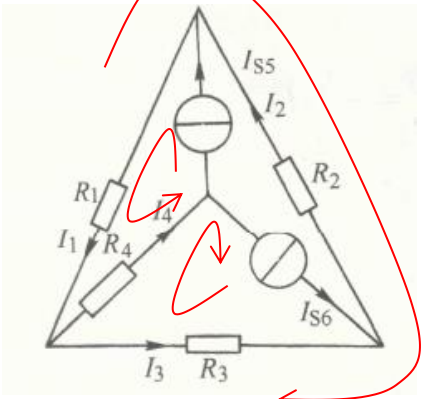
4-- 4.15

列写支路或回路方程



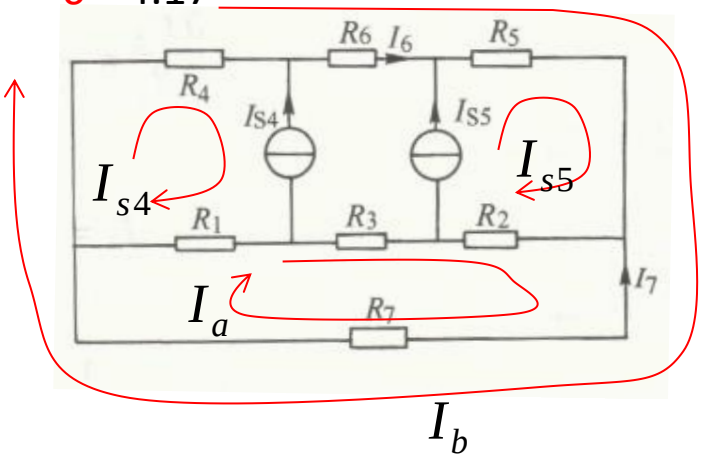
$$(R_4+R_5+R_6)I_a-I_{S7}R_4-(-I_{S8})R_5-I_{S9}R_6=0$$

5-- 4.16



$$(R_1+R_2+R_3)I_a-I_{S5}R_1+I_{S6}R_3=0$$

6-- 4.17

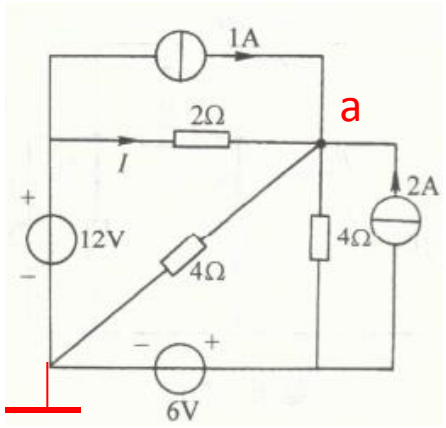


$$(R_1+R_2+R_3+R_7)I_a-I_{S4}R_1-I_{S5}R_2+I_bR_7=0$$

$$(R_4+R_5+R_6+R_7)I_b+I_{S4}R_4+I_{S5}R_5+I_aR_7=0$$

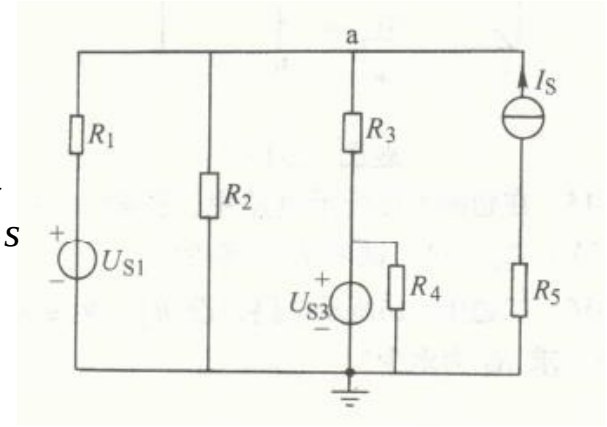
列写节点电压方程

7-- 4.18



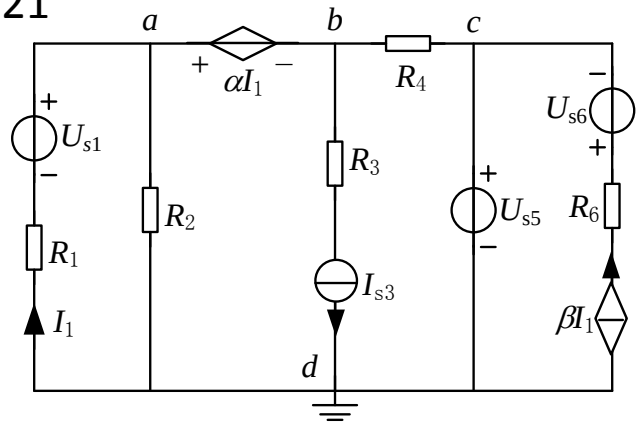
$$\frac{U_a-12}{2}+\frac{U_a}{4}+\frac{U_a-6}{4}=1+2$$

8-- 4.19



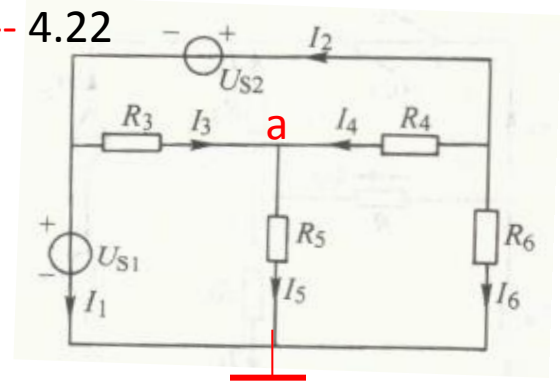
$$\frac{U_a-U_{s1}}{R_1}+\frac{U_a}{R_2}+\frac{U_a-U_{s3}}{R_3}=I_s$$

9-- 4.21



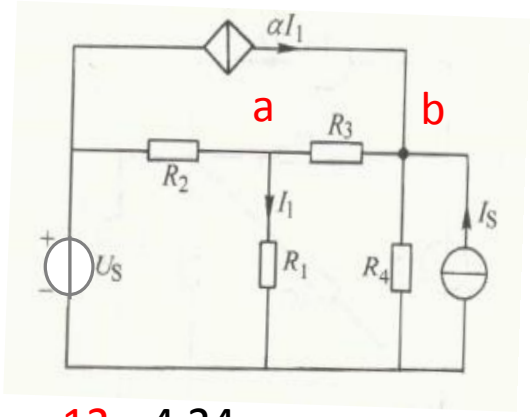
$$\begin{cases}\dfrac{U_a-U_{s1}}{R_1}+\dfrac{U_a}{R_2}+\dfrac{U_a-\alpha I_1-U_{s5}}{R_4}=-I_{s3}\\ U_a=U_{s1}-R_1I_1\end{cases}$$

10-- 4.22



$$\frac{U_{s1}-U_a}{R_3}-\frac{U_a}{R_5}+\frac{U_{s1}+U_{s2}-U_a}{R_4}=0$$

11-- 4.23 选择合适的方程方法

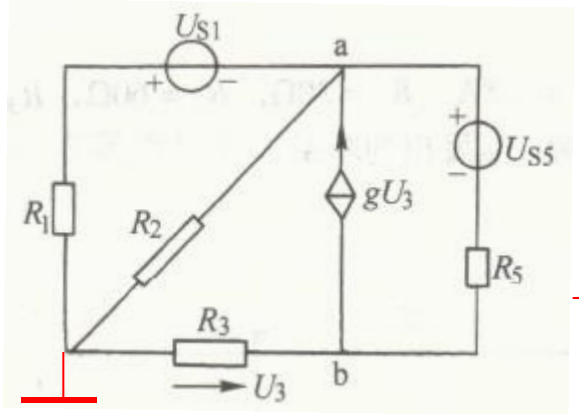


$$\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_3}\right)U_a-\frac{1}{R_2}U_s-\frac{1}{R_3}U_b=0$$

$$\left(\frac{1}{R_4}+\frac{1}{R_3}\right)U_b-\frac{1}{R_3}U_a=I_s+\alpha I_1$$

$$I_1=\frac{U_a}{R_1}$$

12-- 4.24

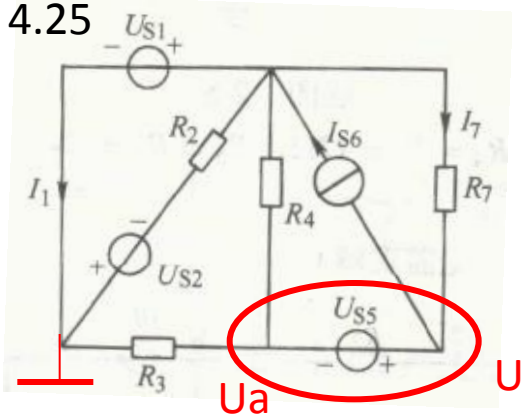


$$\left(\frac{1}{R_1}+\frac{1}{R_2}+\frac{1}{R_5}\right)U_a-\frac{1}{R_5}U_b=-\frac{U_{s1}}{R_1}+\frac{U_{s5}}{R_5}+gU_3$$

$$-\frac{1}{R_5}U_a+\left(\frac{1}{R_5}+\frac{1}{R_3}\right)U_b=-\frac{U_{s5}}{R_5}-gU_3$$

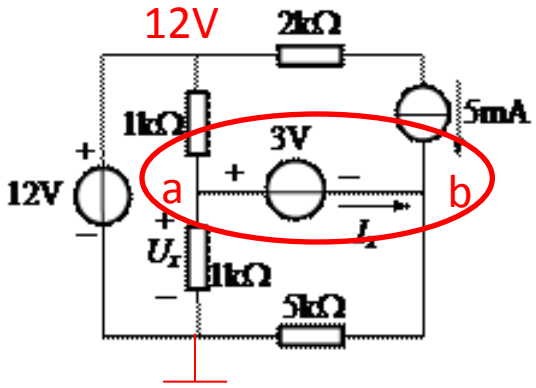
$$U_3=-U_b$$

13-- 4.25



$$\frac{U_a-U_{s1}}{R_4}+\frac{U_a}{R_5}+\frac{U_a-U_{s5}-U_{s1}}{R_7}=-I_{s6}$$

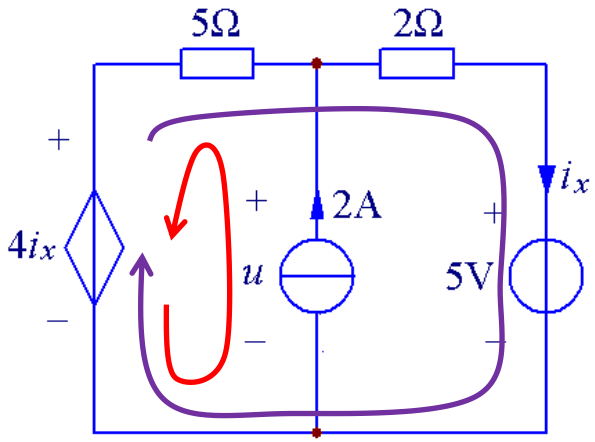
14．用节点法求图示电路中的U_x、I_x　　Ans: U_X=8V，I_X = -4mA



$$\frac{12-U_a}{1k}+5mA=\frac{U_a-0}{1k}+\frac{U_b-0}{5k}$$

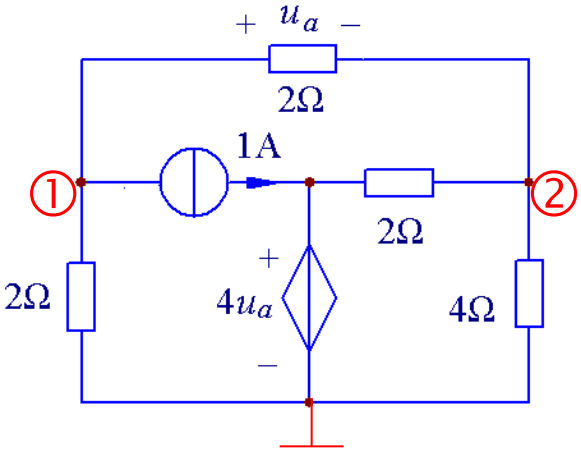
$$U_a-U_b=3$$

15．用 尽可能少的方程求i_x



$$5(i_x-2)+2i_x+5-4i_x=0$$

16．用尽可能少的方程求U_a



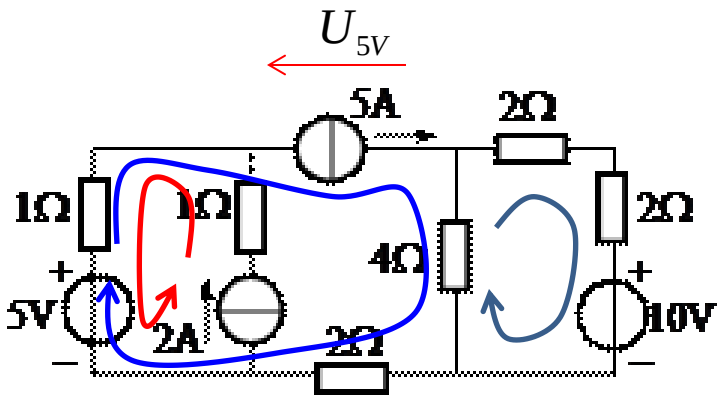
$$\left(\frac{1}{2}+\frac{1}{2}\right)U_1-\frac{1}{2}U_2=-1$$

$$\left(\frac{1}{2}+\frac{1}{2}+\frac{1}{4}\right)U_2-\frac{1}{2}U_1-\frac{1}{2}4U_a=0$$

$$U_a=U_1-U_2$$

$$U_a=-2\left(\frac{U_1}{2}+1\right)$$

17. 最小工作量的方程法求各电源发出的功率（P_{5V}=15W； P_{2A}=8W； P_{5A}=115W； P_{10V}=-12.5）

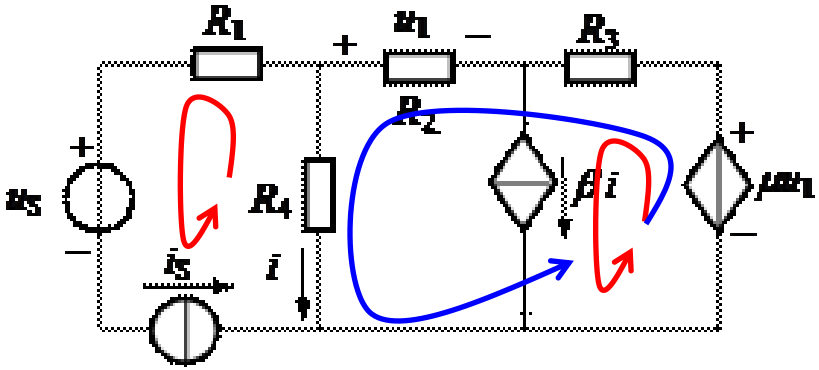


$$4i_x + 10 + 4(i_x - 5) = 0$$

$$i_x = 1.25A$$

$$U_{5V} = 4(5 - 1.25) + 5 \times 2 - 5 + 1 \times 3 = 23V$$

18. 用最小工作量的方程法求u1



$$R_3(i_x + \beta i) + R_2 i_x + R_4(i_x - i_s) = \mu R_2 i_x$$

$$i = (i_x - i_s)$$